

Xiangbo Gao

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Research/Engineering Intern: Autonomous Driving · Perception · Planning

EDUCATION

BS, University of California, Irvine · Computer Science, Mathematics (GPA-3.720) 2018.9 ~ 2023.3
Summer Session, University of California, Berkeley (GPA-3.566) 2019.6 ~ 2019.9
MS, University of Michigan · Robotics plan to start at 2023.9

SKILLS & Research Interest

Programming Language: Python, C/C++

Software: PyTorch, OpenMMLab, Carla Simulator, RLib, TorchGeometry

Research Interests: Autonomous Driving, Perception, Deep Learning, Computer Vision, Motion Prediction, and Planning

PUBLICATIONS

1. **X Gao**¹, et al. "Scale-free and Task-agnostic Attack: Generating Photo-realistic Adversarial Patterns with Patch Quilting Generator" arXiv:2208.06222, 2022
2. M Liu, X Li, **X Gao**³, J Chen, L Shen, & H Wu "Sample hardness based gradient loss for long-tailed cervical cell detection", International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 109-119. Springer, Cham, 2022

RESEARCH EXPERIENCE & LEADERSHIP

University of California, Irvine, USA

Auto-generated Graphical Model in the Autonomous Driving System 2022.5 ~ 2022.11

- Designed a multi-domain autonomous driving dataset for various driving scenarios using the CARLA simulator.
- Developed a probabilistic LSTM structure that encodes states as variational embeddings using Pytorch.

Contrastive Learning in Deep Reinforcement Learning 2022.2 ~ 2022.6

- Proposed a Spring Loss that uses the idea of contrastive learning to align the embeddings in the linear distance.
- Visualized the learned embedding using PCA, which shows better alignment with real-world trajectories.

ZerO Waste Anteaters 2020.9 ~ 2022.4

- Led a team of 8 members to explore technology-driven autonomous waste recycling solutions, including the development and deployment of waste image classification and object detection.
- Inspiring students' interest and guiding them to explore AI, computer vision, and sustainability.

Computer Vision Institute, Shenzhen University, China

Adversarial Attack with Semantic Pattern 2021.4 ~ 2022.10

- Proposed a novel PQ-Attack method that generates scale-free adversarial perturbation and deliveries state-of-the-art attack strength and robustness.

Long-tailed Cervical Cell Detection 2021.5 ~ 2022.1

- Assisted to propose and implement Grad-Libra Loss, achieving 7.8% higher mAP than CE loss on the TCT WSI dataset.

INTERNS & EMPLOYMENTS

Anhui Cowa ROBOT Co., Ltd, Shanghai, China 2023.4 ~ 2023.7

Position: Perception Algorithms Researcher

- Online HD Map Construction with Flow Map Prior (3.2% higher mAP than the baseline)
- Motion Prediction with Historical Trajectories Clustering

Tandll Investment Management, Ltd, Shenzhen, China 2020.6 ~ 2020.8

Position: Full-stack software developer

- Developed a website for management that is currently in use.

Calit 2, University of California, Irvine 2019.2 ~ 2019.7

Position: Software developer

- Developed a VR teaching tool for MA6 Mask Aligner.

COMPETITIONS

[Rank 13th] **CVPR Camera-based online HD map construction challenge 2023** 2022.5

[Rank 1st] **UCI Machine Learning Hackathon**, University of California, Irvine, CA 2020.4

Geometric Pose Affordance: predicting 3D pose from 2D data and light tracing information

[Rank 2nd] **Google Hash Code 2020 Algorithms Competition** 2020.2